

Extension

Sometimes increasing the length of a code by one bit gives a new code with improved error detection or correction, which justifies the (slightly) lower rate.

5.7.5 Definition Let C be a linear code of length n . If $c \in C$, let c^* be the word of length $n + 1$ formed by adding one extra bit to c to make $\|c^*\|$ even. Let $C^* = \{c^* \mid c \in C\}$. This is a code of length $n + 1$, called the *extended code of C* .

5.7.8 Theorem If a linear code C has a $k \times n$ generating matrix G then the extended code C^* has a $k \times (n + 1)$ generating matrix

$$G^* = \begin{pmatrix} G & b \end{pmatrix},$$

where b is a $k \times 1$ column vector such that each row of G^* has even weight.

5.7.9 Theorem If H is a parity check matrix for C , then a parity check matrix for C^* is given by

$$H^* = \begin{pmatrix} H & 1^T \\ 0 & 1 \end{pmatrix},$$

where 1^T is a column vector containing all 1s, and $(0 \ 1)$ is a single extra row.

Example Let C be the $(4, 2, 1)$ -code given by
 $C = \{0000, 1000, 0111, 1111\}$

Let C be an (n, k, δ) linear code, and let C^* have parameters (n^*, k^*, δ^*) . Clearly $n^* = n + 1$ and $k^* = k$. Thus, the rate r^* of C^* is $r^* = k^*/n^* = k/(n + 1)$. Notice that C^* is a slightly less efficient code than C since

$$r = \frac{k}{n} > \frac{k}{n + 1} = r^*.$$

$$\delta^* = \|c^*\| = \begin{cases} \delta, & \text{if } \delta \text{ is even,} \\ \delta + 1, & \text{if } \delta \text{ is odd.} \end{cases}$$

Thus an extended code is only useful when the distance of the original code is odd, in which case the extended code corrects no more errors than the original code, but it detects one more error.

5.7.11 Theorem Let C be a linear (n, k, δ) -code in which δ is odd. Then there exists a linear $(n + 1, k, \delta + 1)$ -code.

Note that Theorems 5.7.3 and 5.7.11 give us the result that a linear (n, k, δ) -code with δ odd exists if and only if a linear $(n + 1, k, \delta + 1)$ -code exists.

Table of general binary codes: <https://aeb.win.tue.nl/codes/binary-1.html>
gives max # of codewords for given n and even δ .

Residual

Let C be a linear (n, k, δ) -code and let $v \in C$ be a codeword of weight δ . By considering equivalent codes, we can assume that v has 1 in the first δ positions, and 0 in the remaining $n - \delta$ positions. Choose a generating matrix G for code C , such that G has v as its first row. Delete the first δ columns of G as well as the row of zeros that results, leaving a $(k - 1) \times (n - \delta)$ matrix. The residual code R of C is the code with this matrix as its generating matrix.

5.7.12 Theorem Let C be a linear (n, k, δ) -code. The residual code is a linear $(n - \delta, k - 1, \lceil \frac{\delta}{2} \rceil)$ -code.

This construction was used in the proof of the Griesmer bound.

$(a|a+b)$ Construction

Let C_1 be a linear (n, k_1, δ_1) -code and let C_2 be a linear (n, k_2, δ_2) -code. For each $c_1 \in C_1$ and $c_2 \in C_2$, we construct a word of length $2n$ whose first n bits are the word c_1 and whose last n bits are the word $c_1 + c_2$. These words of length $2n$ are a new linear code and we refer to this construction the $(a|a+b)$ construction. Note that $a \in C_1$ and $b \in C_2$.

Example:

Let C be the linear $(4, 2, 1)$ -code, $C = \{0000, 1000, 0111, 1111\}$

Let D be the linear $(4, 2, 2)$ -code, $D = \{0000, 1100, 0011, 1111\}$

The $(a|a+b)$ construction with $a \in C$ and $b \in D$ gives:

0000 0000	1000 1000	0111 0111	1111 1111
0000 1100	1000 0100	0111 1011	1111 0011
0000 0011	1000 1011	0111 0100	1111 1100
0000 1111	1000 0111	0111 1000	1111 0000

$n = 8$

$k = 4$

$\delta = 2$

The $(a|a+b)$ construction with $a \in D$ and $b \in C$ gives:

0000 0000	1100 1100	0011 0011	1111 1111
0000 1000	1100 0100	0011 1011	1111 0111
0000 0111	1100 1011	0011 0100	1111 1000
0000 1111	1100 0011	0011 1100	1111 0000

$n = 8$

$k = 4$

$\delta = 1$

Let C_1 be a linear (n, k_1, δ_1) -code with generating matrix G_{C_1} and parity check matrix H_{C_1} , let C_2 be a linear (n, k_2, δ_2) -code with generating matrix G_{C_2} and parity check matrix H_{C_2} , and let C be the code constructed from C_1 and C_2 using the $(a|a+b)$ construction with $a \in C_1$ and $b \in C_2$. Then G_C and H_C are a generating matrix and parity check matrix, respectively, for C , where

$$G_C = \begin{pmatrix} G_{C_1} & G_{C_1} \\ 0 & G_{C_2} \end{pmatrix} \quad H_C = \begin{pmatrix} H_{C_1} & H_{C_2} \\ 0 & H_{C_2} \end{pmatrix}.$$

5.7.14 Theorem Let C_1 be a linear (n, k_1, δ_1) -code and let C_2 be a linear (n, k_2, δ_2) -code. Then there exists a linear $(2n, k_1 + k_2, d)$ -code where $d = \min\{2\delta_1, \delta_2\}$.

Proof idea for Theorem 5.7.14

Length: 'a' has length n , 'a+b' has length n , so a|a+b has length $2n$.

Dimension: For each codeword of C_1 , we get $|C_2|$ new codewords, so # new codewords $= 2^{k_1} \cdot 2^{k_2} = 2^{k_1+k_2}$, so dimension is $k_1 + k_2$.

Distance: Let u be a word of weight δ_1 in C_1 . Then $u|u$ is a new codeword (since $0 \in C_2$) and $\|u|u\| = 2\delta_1$ so the distance of the new code is $\leq 2\delta_1$.

If $v \neq 0$ and $v \in C_2$, then $\|a|a+v\| \geq \|v\| \geq \delta_2$

Since if bit i of $v = 1$ then:

Case 1 bit i of $a+v$ is 0 if bit i of a is 1.

Case 2 bit i of $a+v$ is 1 if bit i of a is 0.

So the new code has distance $\min\{2\delta_1, \delta_2\}$

Exercise

Let C_1 be the $(7, 4, 3)$ -Hamming code and C_2 be the $(7, 1, 7)$ -code $\{0000000, 1111111\}$. What are the parameters that result from the $(a|a+b)$ construction if

(a) $a \in C_1$ and $b \in C_2$.

(b) $a \in C_2$ and $b \in C_1$.

Summary of techniques:

Shortening: $(n, k, \delta) \rightarrow (n-1, k-1, \delta)$

Projection: (n, k, δ) with $\delta > 1 \rightarrow (n-1, k, \delta-1)$

Extension: (n, k, δ) with δ odd $\Rightarrow (n+1, k, \delta+1)$

$\exists (n, k, \delta)$ with δ odd iff $\exists (n+1, k, \delta+1)$

Residual: $(n, k, \delta) \rightarrow (n-\delta, k-1, \lceil \frac{\delta}{2} \rceil)$

$(a|a+b)$: $(n, k_1, \delta_1) + (n, k_2, \delta_2) \rightarrow (2n, k_1+k_2, \min\{2\delta_1, \delta_2\})$

6.1. Reed-Muller Codes

Mariner 9 was successfully launched on 30 May, 1971, and became the first artificial satellite of Mars when it arrived on 13 November, 1971 and went into orbit. It transmitted pictures back to Earth for nearly a year and completed its final transmission on 27 October, 1972.

Mariner 9 had a radio transmitter with a power of only 20 watts, and was transmitting over a distance of 84 million miles. Despite this, near-perfect pictures were obtained. Each picture transmitted by Mariner 9 is made up of over half a million tiny picture elements forming a rectangular array. Each picture element is a uniform shade of grey, the precise shade being specified by a 9-bit binary number. Thus each picture was represented using 5,250,000 bits, in grey-scale. Given the large distances, low power, poor reliability of electronics in the transmitter and receiver, and the general background noise of space, many errors occurred in transmission. Hence it was necessary to perform extensive error correction. Each message was divided into packets of 6 bits, and then each string of 6 bits was encoded into a 32 bit word. Thus a picture represented by 5 and a quarter million bits, was transmitted using over 5 times that number of bits.

The code used by the Mariner 9 was $RM(5)$, which has 64 codewords, each of length 32. Having distance 16, it can correct any received word containing 7 or fewer errors.

a first-order Reed-Muller code,
a linear $(32, 6, 16)$ -code.

$\frac{7}{32} = 0.21875$, so errors in $\approx 20\%$ of any word was ok.

6.1.3 Definition The first-order Reed-Muller code of length 2^m , denoted $RM(m)$, is the linear code given by the generating matrix G_m whose columns are the binary representations of the integers from 2^m to $2^{m+1} - 1$.

Recall that the *binary representation* of a non-negative integer x is the binary sequence $x_t x_{t-1} \dots x_2 x_1 x_0$ such that

$$x = x_t \cdot 2^t + x_{t-1} \cdot 2^{t-1} + \dots + x_2 \cdot 2^2 + x_1 \cdot 2^1 + x_0 \cdot 2^0.$$

$$\begin{array}{c}
 \text{RM}(1) \\
 G = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}
 \end{array}
 \quad
 \begin{array}{c}
 \text{RM}(2) \\
 G = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}
 \end{array}
 \quad
 \begin{array}{c}
 \text{RM}(3) \\
 G = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}
 \end{array}$$

$$\begin{array}{c}
 \text{RM}(4) \\
 G_4 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}
 \end{array}$$

Exercise: Show that $RM(3)$ can be constructed from $RM(2)$ (and another suitable code) using the $(a|a+b)$ construction.

$(a|a+b)$ Construction

Let C_1 be a linear (n, k_1, δ_1) -code and let C_2 be a linear (n, k_2, δ_2) -code. For each $c_1 \in C_1$ and $c_2 \in C_2$, we construct a word of length $2n$ whose first n bits are the word c_1 and whose last n bits are the word $c_1 + c_2$. These words of length $2n$ are a new linear code and we refer to this construction the $(a|a+b)$ construction. Note that $a \in C_1$ and $b \in C_2$.

6.1.5 Theorem $RM(m)$ is a $(2^m, m + 1, 2^{m-1})$ -code.

Proof Since G_m has $(2^{m+1} - 1) - (2^m - 1) = 2^m$ columns, the code $RM(m)$ has length 2^m . The number of bits in the binary representation of $2^{m+1} - 1$ is $m + 1$, so G_m has $m + 1$ rows, which are easily seen to be linearly independent. Thus, $RM(m)$ has dimension $m + 1$. Clearly, $RM(m)$ contains 0_{2^m} and 1_{2^m} , and it can be shown that every other codeword in $RM(m)$ has weight 2^{m-1} . Thus, $RM(m)$ has distance 2^{m-1} . $\square$

Reed-Muller codes are useful on noisy channels as their distance is large for their length. There are also fast encoding and decoding methods for the first-order Reed-Muller codes.

6.2. Hamming Codes

6.2.1 Definition For $r \geq 2$ let $H = H_r$ denote a $(2^r - 1) \times r$ matrix whose rows consist of all non-zero vectors of length r , each occurring exactly once. The corresponding linear code is called a *Hamming code* of length $2^r - 1$.

$r=2$

$$H_2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$A(3, 1, 3)$ -code

$r=3$

$$H_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$A(7, 4, 3)$ -code

$r=4$

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$A(15, 11, 3)$ -code.

Thus, for all $r \geq 2$ the Hamming code with parity check matrix $H = H_r$ is a $(2^r - 1, 2^r - 1 - r, 3)$ linear code.

Since the number of cosets is $2^{n-k} = 2^r$ and there are 2^r error patterns of weight 0 or 1, the SDA for a Hamming code is easy to construct.

coset leader	syndrome
000...0	000...0
I_{2^r-1}	H

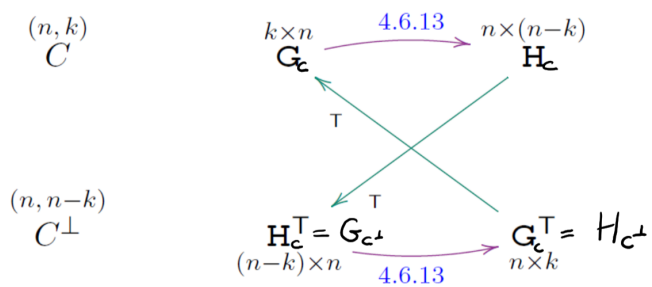
Exercise: The following generating matrix for a $(7, 4, 3)$ -Hamming code was used to encode message words of length 4 and the resulting codewords were transmitted over a BSC. Determine the most likely message word corresponding to the received word 1010100.

$$G = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Since Hamming codes have an odd distance, it is sometimes useful to consider the extended Hamming codes. For $r \geq 2$, the extended Hamming code is a $(2^r, 2^r - 1 - r, 4)$ code.

Code	Generating Matrix	Parity Check Matrix
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5.7.9 Theorem If H is a parity check matrix for C , then a parity check matrix for C^* is given by



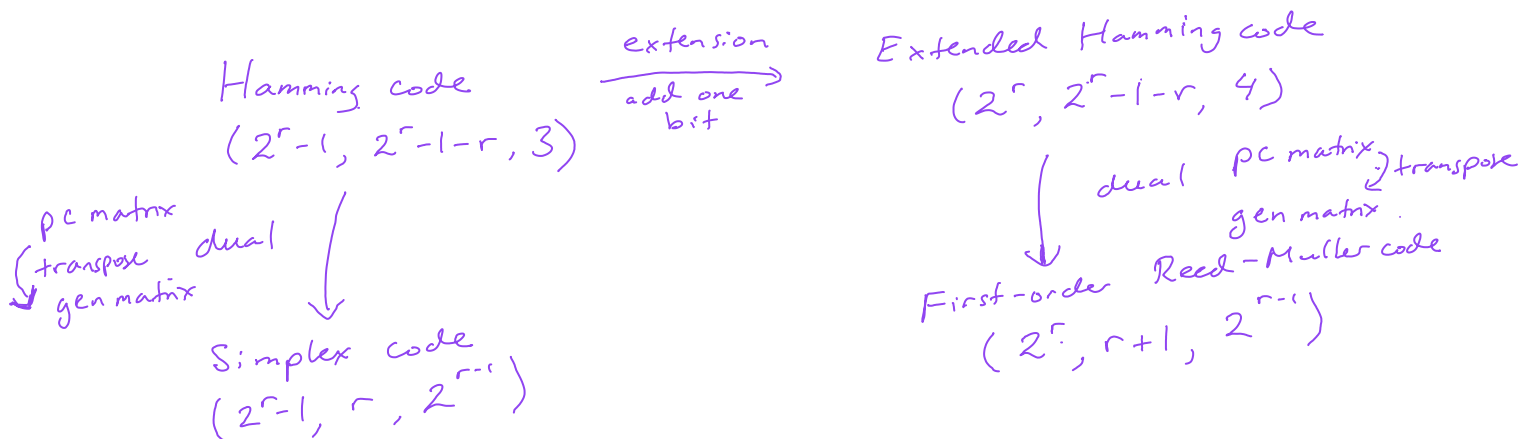
$$H^* = \begin{pmatrix} H & 1^T \\ 0 & 1 \end{pmatrix},$$

Exercise Show that the dual of the extended Hamming code of length 8 is equivalent to $RM(3)$.

Recall that the rows of the parity check matrix for a $(2^r-1, 2^r-1-r, 3)$ -Hamming code are all the non-zero words of length r (so the binary representations of $1, 2, \dots, 2^r-1$).

To build a parity check matrix for the extended code we add a row of zeros (so now we have binary representations of the numbers $0, 1, 2, \dots, 2^r-1$) and a column of I 's. Where you place the column of I 's doesn't matter, so if you place it as the first column, you have that the rows are the binary representations of the numbers $2^r, 2^r+1, \dots, 2^{r+1}-1$. Take the transpose and you have a generating matrix for $RM(r)$.

6.2.3 Definition The dual of the Hamming code of length $2^r - 1$ is called the simplex code of length $2^r - 1$.



Recall various families of binary linear codes we have seen

$(n, n, 1)$ -codes all words in $\mathbb{F}_2^n$ (perfect codes)

$(n, 1, n)$ -codes $\{\underline{0}, \underline{1}\}$ (perfect for n odd)

$(2^m, m+1, 2^{m-1})$ -codes 1st order Reed-Muller codes $RM(m)$

$(2^r-1, 2^r-1-r, 3)$ -codes Hamming codes (perfect codes)

$(2^r-1, r, 2^{r-1})$ -codes Simplex codes (dual of Hamming codes).

Use the $(a|a+b)$ construction to show that there exists a

① $(30, 15, 6)$ -code

② $(2^{r+1}-2, r+1, 2^r-1)$ -code

Question [Todd Ebert (1998)] Suppose n players enter a room and a red or blue hat is placed on each person's head. The colour of each hat is determined randomly. Each person can see all the other players' hats but not their own.

No communication is allowed. Once they have had a chance to look at the other hats, the players must each simultaneously guess the colour of their own hat, or pass.

The group wins \$1 billion if at least one player guesses correctly and no players guess incorrectly.

What is the optimal strategy? *Note that the players decide on a strategy before the game starts.*